

HAT-P-19b

R-1418

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_230919 · created 2026-08-01T06:00:49+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460206.7341 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.011
Transit depth (Rp/Rs)^2	1.97 +/- 0.3 [%]
Orbital Inclination (inc)	88.69 +/- 0.68
Airmass coefficient 1 (a1)	1.008 +/- 0.013
Airmass coefficient 2 (a2)	-0.0053 +/- 0.0077
Scatter in the residuals of the lightcurve fit is	1.3 %
Best Comparison Star	#3 - [263, 221]
Optimal Aperture	3.98
Optimal Annulus	14.9
Transit Duration (day)	0.1155 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.14 $\pm$ 0.011 vs 0.1418 (deviation 0.16 σ)
Duration fit vs published	0.1155 d vs 0.1196 d (deviation 0.76 σ)
Mid-time O-C	-12.2 min
Depth SNR	12.7
Residual scatter	1.3 %

Light curve

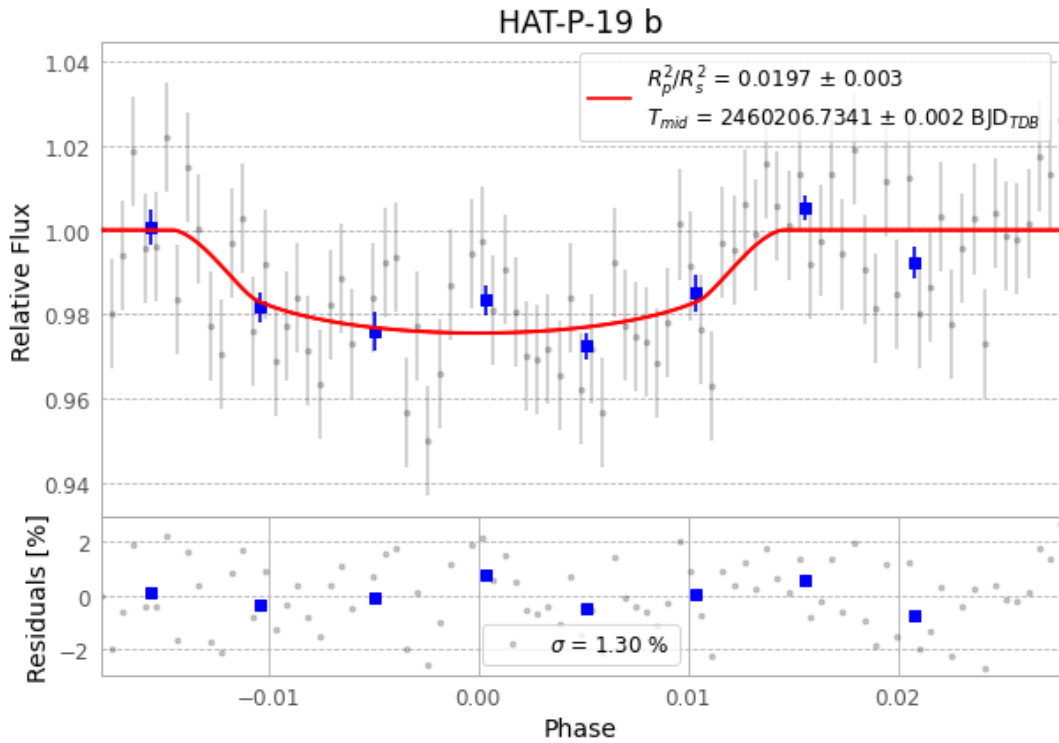


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-09-18.png (SHA-256 fe9af909823a76e0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 838d22317598e742...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), HATP-19230919034515.FITS → HATP-19230919080915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-230919.log (3ac515a972dc...), FinalLightCurve_HAT-P-19 b_2023-09-18.png (fe9af909823a...), FinalLightCurve_HAT-P-19 b_2023-09-18.csv (6f2242ac64fc...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 06:00Z.